

Updating Firmware

This guide applies if you encounter the following error message:

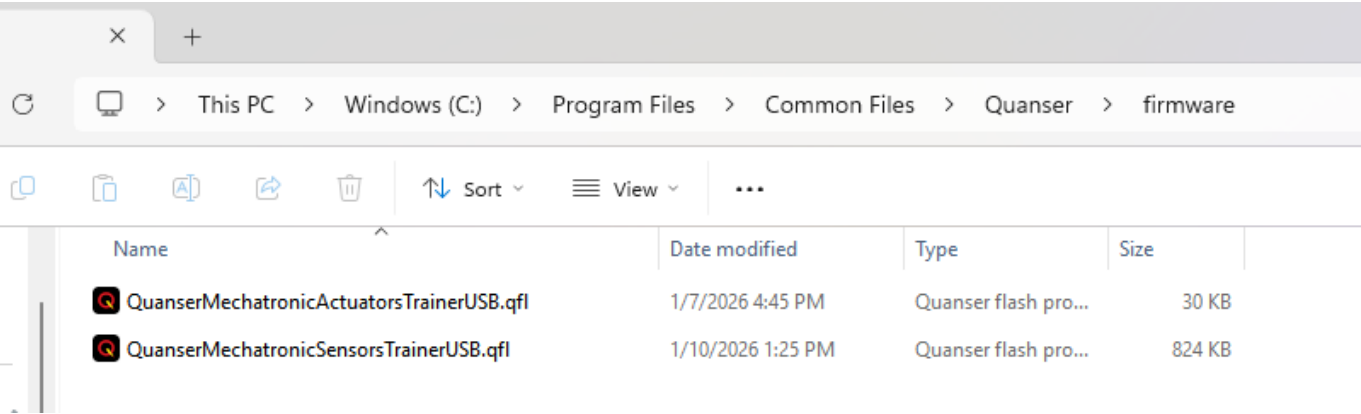
Version information retrieved from the HIL board does not match the driver. It may be necessary to update the QUARC software to get the latest drivers.

Students should report this error to their lab supervisor or professor before proceeding, as this procedure should only be performed by authorized personnel.

To start, please verify that the following folder exists on your system:

C:\Program Files\Common Files\Quanser\firmware

The folder should contain the files shown below:



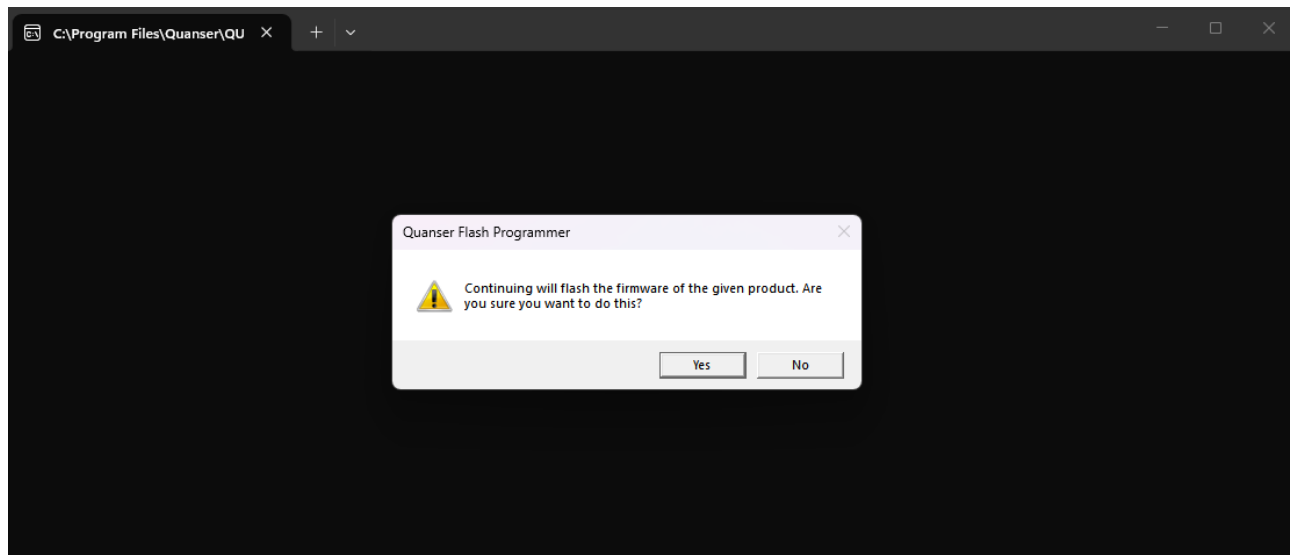
If you do not have this folder or the required firmware files, see [Getting the Correct Firmware](#).

If you do have the folder, proceed with this guide.

Updating the Firmware in the Board

Ensure that only one Quanser Mechatronic device is connected to your PC. Do not disconnect the device throughout the entire process.

1. Open the folder that contains the firmware at C:\Program Files\Common Files\Quanser\firmware. The folder should contain the files shown above.
2. Double-click the appropriate file for your device (Sensors or Actuators).
3. A dialog box and command terminal window will appear. Select 'Yes' to proceed.



4. The command terminal will display status information as it identifies and programs the device.

```
C:\Program Files\Quanser\QU x + v

Program file:      C:\Program Files\Common Files\Quanser\firmware\QuanserMechatronicSensorsTrainerUSB.qfl
Program file format: qfl
Format version:    2
Vendor ID:         0x21df
Product ID:        0x0027
Min. PCB revision: 5
Max. PCB revision: 6
Minimum address:   0x0000000000020000
Maximum address:   0x000000000001ffffff
Preserve address:  0x0000000000000000
Preserve length:   0
Bootloader vendor ID: 0x21df
Bootloader product ID: 0x0028
Alignment:         32
Bootloader version: 1
Protocol:          USB
Flags:             QFU
Number of devices: 1

Re-enumerating devices entering bootloader mode. Please wait...
|
```

5. Upon successful completion of the process, which typically takes approximately 30 seconds to a minute, the following screen will be displayed.

```
C:\Program Files\Quanser\QU x + v

Program file format: qfl
Format version:    2
Vendor ID:         0x21df
Product ID:        0x0027
Min. PCB revision: 5
Max. PCB revision: 6
Minimum address:   0x0000000000020000
Maximum address:   0x000000000001ffffff
Preserve address:  0x0000000000000000
Preserve length:   0
Bootloader vendor ID: 0x21df
Bootloader product ID: 0x0028
Alignment:         32
Bootloader version: 1
Protocol:          USB
Flags:             QFU
Number of devices: 1

Re-enumerating devices entering bootloader mode. Please wait...
Bootloader version: 1
PCB version: 6
Unlocking memory...Memory unlocked.
Erasing memory...Memory erased.
Writing memory.....Memory written.
Locking memory...Memory re-locked.
Verifying memory.....Memory verified.
Firmware flashed successfully on device 0.
Please cycle the power on the device.
Press Enter to exit.
|
```

6. Disconnect and reconnect the device to your computer. Device testing should now proceed without errors.

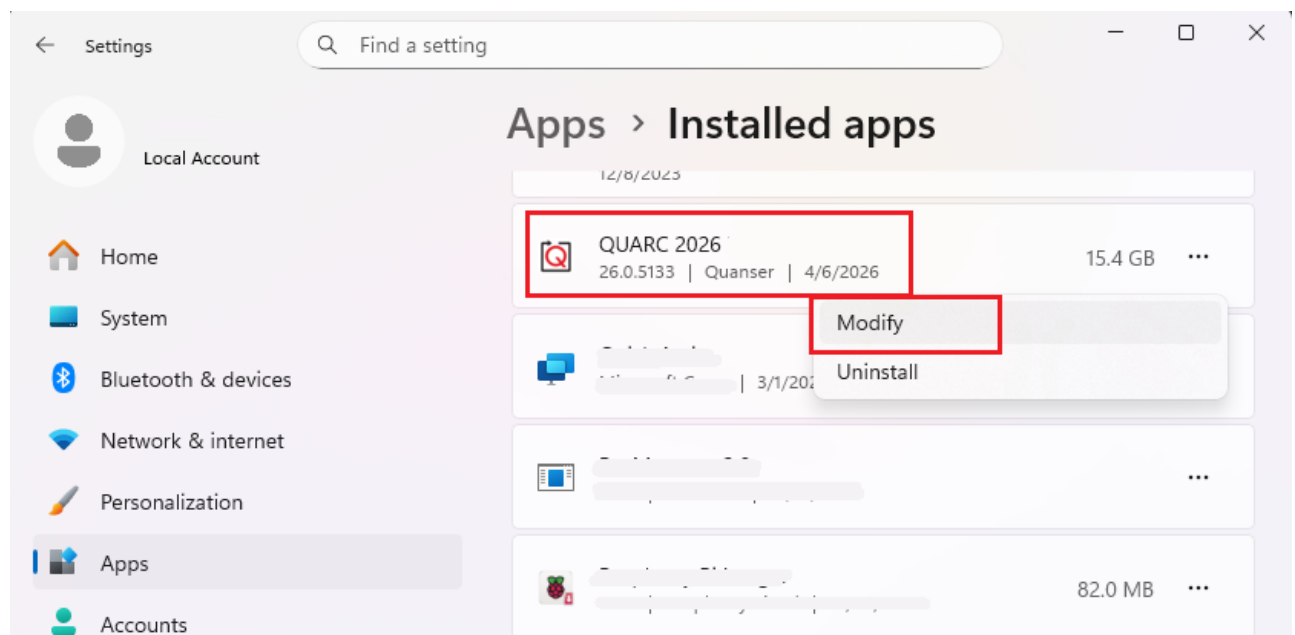
If the issue persists, please contact Quanser Technical Support at tech@quanser.com.

Getting the Correct Firmware

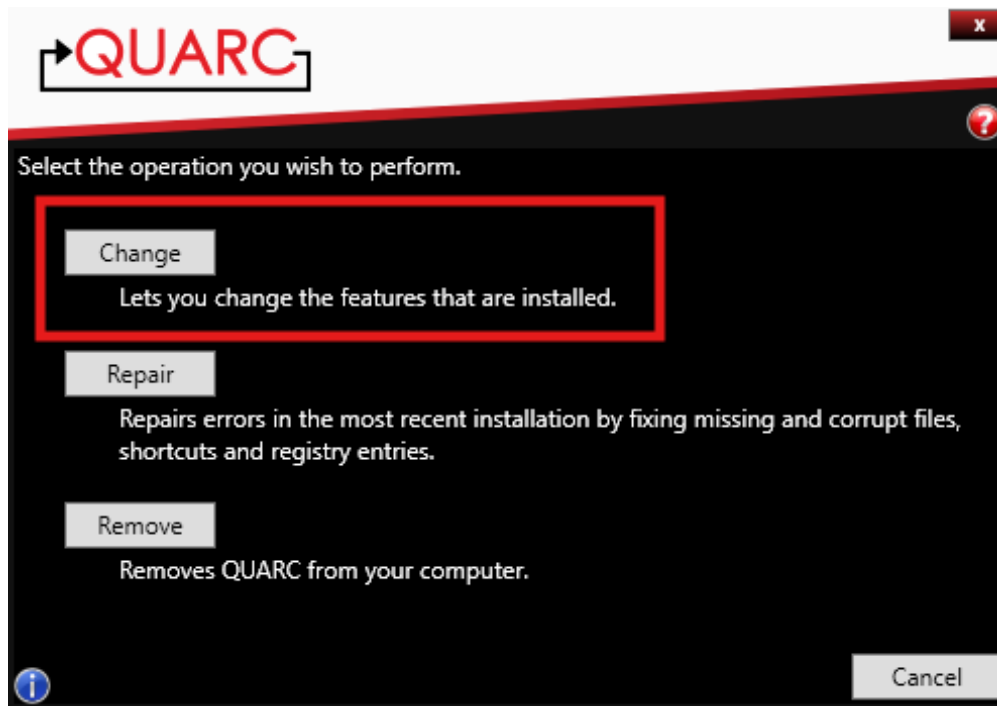
This section only applies if you did not find the firmware in your system.

If the firmware folder is not present on your system, this is typically because QUARC is installed without the Firmware package.

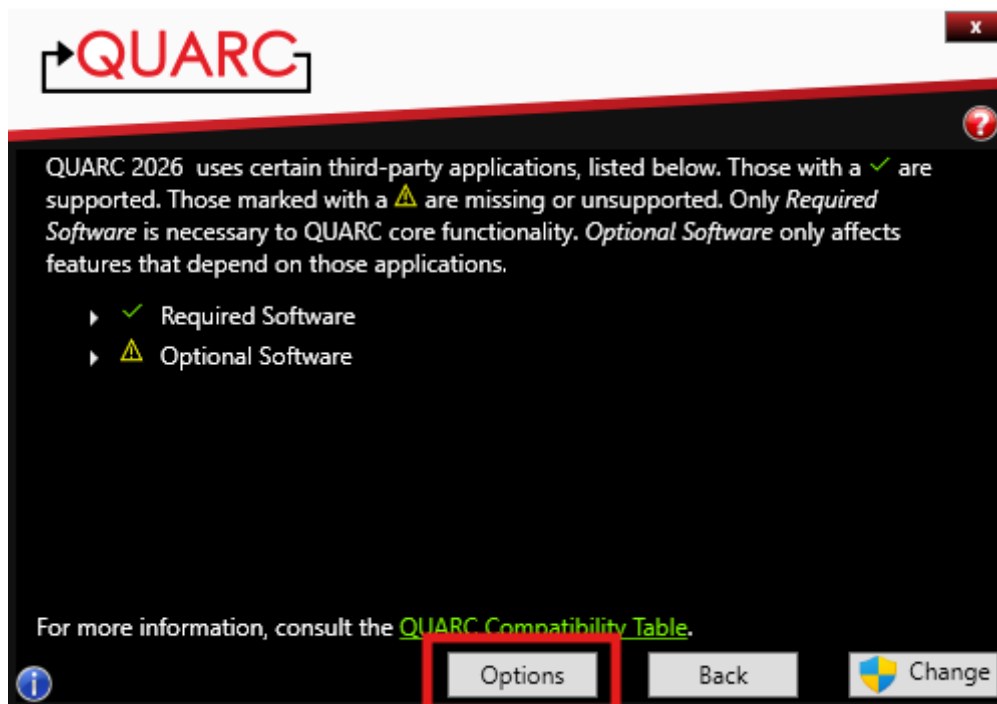
1. To resolve this, navigate to **Apps > Installed apps** in your Windows settings. Locate your QUARC installation and select **Modify** from the options menu (three dots) as shown below.



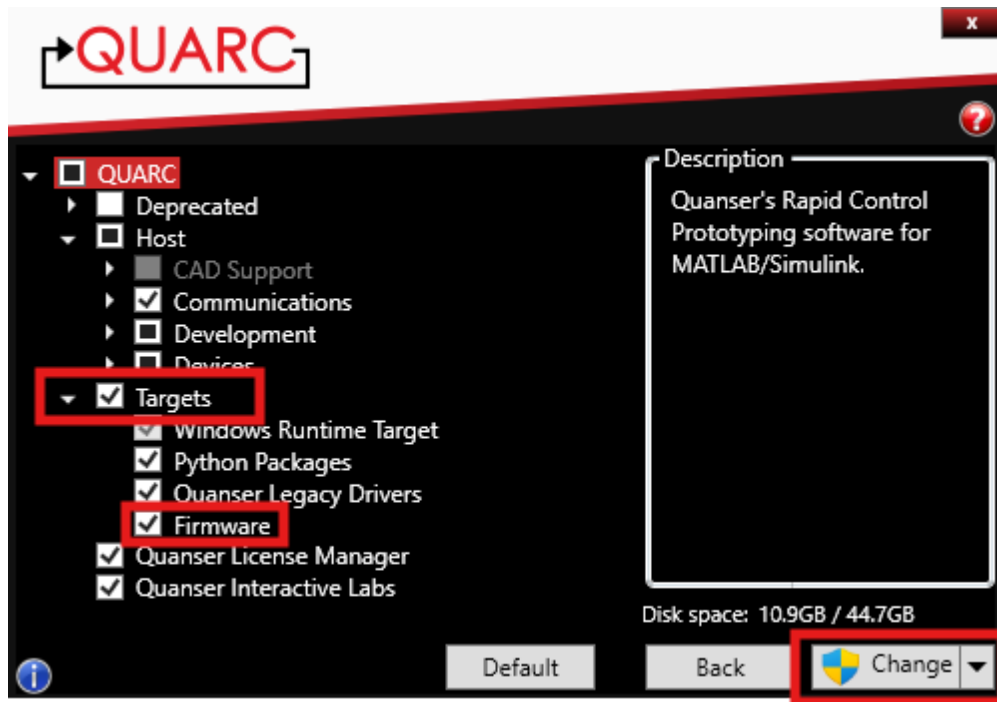
2. After accepting the dialog box, the following window will appear. Select **Change** to modify your installation.



3. Select **Next** until you see the page below. Select **Options** to modify the components included with your installation.



4. Locate the Targets dropdown and expand it to view the available options. Select Firmware and enable it by verifying that the checkbox next to it is enabled.



5. Once selected, select **Change** to begin the update. A progress bar will display during the modification process. Upon completion, the following confirmation screen will be displayed. Select **Close**.



6. The firmware folder and associated files will now be available on your system. Proceed with the firmware update process described [in the previous section](#).